



CS 560: Text Retrieval and Mining

Course Code	CS 560
Course Name	Text Retrieval and Mining
L-T-P-C	3-0-0-3
Prerequisites	CS309 Information and Database Systems
Intended for	B Tech, MS and PhD
Distribution	Elective for CS
Approval	7th Senate (Mar 2015)

Course Contents

- **Introduction to IR:** What IR means, what its goals are, what entities it attempts to retrieve, the criteria by which IR systems are evaluated. Web search engine as a case study. [3 Lectures]
- **Introduction to TM:** Structured vs Unstructured Data, Document Classification and Information Retrieval, Clustering and Organizing Documents, Textual Information to Numerical Vectors, Tokenization, Lemmatization, Stemming, Vector generation, Sentence boundary determination, Part of speech tagging, Parsing, Feature generation [8 Lectures]
- **Text Mining Techniques:** Similarity and nearest neighbor methods, Decision rules, Decision Trees, Linear Scoring methods, Key word search, Document matching, Inverted list, Clustering- K-means, Centroid clustering, Hierarchical clustering [8 Lectures]
- **Looking for Information in Documents:** Finding Patterns and entities, Coreference resolution, Relationship Extraction, Template filling and database construction [6 Lectures]
- **Case Studies:** Assigning topics to News Articles, Filtering email (Enron example) [3 Lectures]
- Hands on experience and mini projects

Software R with text mining package will be used for text mining the following datasets:

- Twitter dataset
- Titanic survivor dataset
- Blogging dataset
- Spam training dataset

Projects will be based on the real world applications like (not an exhaustive list)

- Spam filtering
- Fraud detection by investigating notification of claims
- Automatic labeling of documents in business libraries
- Creating suggestion and recommendations (like amazon)
- Monitoring public opinions (for example in blogs or review sites)
- Measuring customer preferences by analyzing qualitative interviews
- Fighting cyberbullying or cybercrime in IM and IRC chat

Reference Books/Material

1. C. Manning, P. Raghavan, and H. Schtze, **Introduction to Information Retrieval**, Cambridge University Press, 2008.
2. Sholom M. Weiss, Nitin Indurkha, Tong Zhang, Editors David Gries Fred B. Schneider, **Fundamentals of Predictive Text Mining**, Springer
3. W. Fan, L. Wallace, S. Rich, Z. Zhang, **Tapping the power of text mining**, Communications of ACM, 49(9), 76-82, 2006.
4. The Journal of Statistical Software article **Text Mining Infrastructure in R**
5. Ingo Feinerer, **Introduction to the tm Package: Text Mining in R**, 2014. (<http://cran.r-project.org/web/packages/tm/vignettes/tm.pdf>)